Thinking, Fast and Slow, Formalized: When a Calibrated Decision Model Should Wake a Reasoning Model in Service and Software Systems

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Abstract

Kahneman's account of two modes of thought — a fast, automatic System 1 and a slow, effortful System 2 that is engaged only when needed — has become an engineering blueprint: "System One" decision models such as Jev return typed, calibrated judgments in half a second without generating text, while hybrid reasoning models such as Qwen3-8B can think at a controllable token budget. We give the interaction between such systems a mathematical form that tracks the psychological literature closely. Conflict detection and override become an exact escalation identity: the value of deliberation equals the rate at which the two systems disagree times System 2's win margin on those disagreements, the first factor observable without labels. Trustworthy intuition (Klein and Kahneman) becomes an act-band error guarantee under calibration, with a finite-sample rule for when calibration fails. Anchoring and rationalisation become an exact paired decomposition. Kahneman's theory of effort as a limited resource becomes an expected-value-of-control problem whose solution produces the familiar act/deliberate/give-up bands for any shape of thinking returns, including the inverse scaling recently observed in reasoning models. The *Noise* programme becomes a theorem that aggregating System-2 samples corrects noise but entrenches bias. Every assumption and result is restated in plain words, and we specify exactly when the reasoning model checks the decision model's answer and when the decision model checks the reasoning model's. We map the results to service triage, software and requirements engineering, logistics, consumer behaviour, e-government and agentic software. An evaluation protocol is pre-specified and frozen together with the code that implements it; it has not yet been run, so no measurements of either model are reported. We close with a staged plan — offline replay, shadow mode, canary — for testing the approach in a live service.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*.

Keywords

dual-process theory, System 1 and System 2, metacognition, LLM cascades, calibration, test-time compute, service systems, software engineering

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1 Introduction

Kahneman's *Thinking, Fast and Slow* [20] popularised a distinction long studied in psychology [11, 37]: judgments arrive quickly and automatically from System 1, and a slower, effortful System 2 is recruited when something signals that the quick answer may be wrong. The distinction was never only descriptive. It carried a theory of *when* to think — effort is costly and limited [19], System 2 is a "lazy controller" that endorses intuition unless alarmed, and the alarm is itself an intuition about one's own confidence [41].

AI systems are now being built to this blueprint. Jev is presented by its developer as the first "System One" model: given a text *state* and typed questions (yes/no, choice, score), it returns probabilities and a calibrated confidence in about half a second and generates no text [17]. Qwen3-8B combines thinking and non-thinking modes in one model and lets the caller cap reasoning with a token budget [33]. Practitioners already pair them — "Jev routes and verifies things, while the LLM provides the language" [31] — in routers, judges and triage services. What is missing is a theory of the pairing that is precise enough to design and audit such services, and faithful enough to the psychology to justify the vocabulary.

This paper supplies one. Each result formalises a specific construct from Kahneman's programme and its successors (Table 4): conflict detection and override [10, 32], the conditions under which intuition can be trusted [22], anchoring and rationalisation [32, 43], effort allocation [19, 27, 36], and noise [23]. We then show how the resulting controller applies to problems central to this conference — service operations, software engineering, logistics, consumer behaviour and e-government — and give a pre-specified protocol for testing it. Because the psychology of dual processes has itself been shaken by failed replications [16], we report no results we could not verify: the authoring environment could reach neither model, so the empirical claims are stated as falsifiable hypotheses rather than findings. Sec. 8.1 states exactly what has and has not been done.

Contributions. (1) A formal correspondence between Kahneman's constructs and quantities in a two-model decision system (Sec. 4). (2) Nine propositions, including an exact escalation identity, a shape-free theory of thinking budgets, and a noise-versus-bias theorem for self-consistency (Sec. 5). (3) Five interaction patterns tied to competing dual-process architectures, with an exact account of when each model checks the other (Sec. 6). (4) Applications and a worked example for service and software systems (Sec. 7). (5) A pre-specified protocol frozen together with its code, and a staged plan for testing the controller in a live service (Secs. 8–9).

What is new and what is borrowed. Several ingredients are textbook statistics, and we say so: Table 1 separates what we take from

Table 1: What is borrowed and what is new.

Element	Borrowed from	New here
Escalation identity (Prop. 5.1)	cascades, deferral [6, 14]	exact split of the value of thinking into a label-free factor (disagreement) and a labelled one (who wins)
Act band (Prop. 5.3, Rem. 1)	selective prediction; Learn-then-Test [2, 14]	no new mathematics; applied to a System-One model's own confidence, read as Kahneman–Klein's validity conditions
Anchoring (Prop. 5.4)	anchoring; advice taking [3, 43]	exact paired split into anchor, reactance and drift, with no model of how the hint acts
Budgets (Props. 5.6–5.8)	EVC; resource rationality; test-time scaling [27, 36, 38]	act/think/give-up bands for *any* return curve; band edges depend on the curve only through $\rho^\star$; minimum effective budget
Fallbacks (Props. 5.9–5.10)	Bayes decisions; conformal sets [35]	no new mathematics; placed as the give-up branch
Voting (Prop. 5.11)	Condorcet; self-consistency [15, 46]	no new mathematics; read as *Noise*'s bias–noise split, with a prediction about people
Patterns P3–P5	SOFAI [4]; LLM cascades [6]	label-free probe; System 1 auditing System 2; blind-then-reconcile
Protocol, test plan	—	hypotheses H1–H9 with falsifiers; code; staged deployment test

prior work from what this paper adds. *Reading guide.* Each result is followed by an "In plain words" paragraph; Table 3 explains the notation and Table 5 lists every assumption, how it is checked and what happens if it fails.

2 Kahneman's Two Systems

2.1 Two modes, one mind

Kahneman describes System 1 as operating "automatically and quickly, with little or no effort and no sense of voluntary control" and System 2 as allocating attention to effortful mental activities [20]. Earlier formulations contrasted associative and rule-based reasoning [37]; Evans and Stanovich [11] later argued that the defining property of Type 2 processing is reliance on working memory and hypothetical thinking, not speed as such. A central mechanism is *attribute substitution*: when a target question is hard, System 1 answers an easier related one [21]. Frederick's Cognitive Reflection Test (CRT) [12] operationalises the resulting failure — the bat-and-ball item elicits a fast, confident, wrong answer that only reflective override corrects.

2.2 Heuristics, confidence and their failures

Tversky and Kahneman's heuristics — representativeness, availability, and anchoring and adjustment — produce systematic biases [43]. Kahneman's "what you see is all there is" (WYSIATI) explains why confidence tracks the coherence of the story System 1 constructs rather than the quality of the evidence, producing an *illusion of validity* [20]. Whether intuitive confidence deserves trust was the subject of Kahneman and Klein's adversarial collaboration: intuitions are trustworthy only in environments with valid cues and adequate opportunity to learn them through feedback [22].

2.3 Metacognition: the alarm and the controller

If System 2 is engaged only when needed, something must decide when. Thompson et al. [41] showed that an initial answer arrives with a *feeling of rightness* whose strength predicts both rethinking time and the probability of changing the answer. De Neys [10] argued that reasoners often *detect* conflict between a heuristic answer and a normative one even when they go on to give the heuristic answer. Pennycook, Fugelsang and Koehler [32] integrated these findings in a three-stage model: initial responses are generated (Stage 1); conflict between them is or is not detected (Stage 2); and if analytic processing is engaged (Stage 3), it either *rationalises* the initial answer or *decouples* from it and overrides. Ackerman and Thompson [1] frame the whole as meta-reasoning — monitoring that feeds *control*, including the decision to stop thinking or give up.

2.4 Effort as a resource

Before the two-systems vocabulary, Kahneman's *Attention and Effort* [19] treated mental effort as a limited capacity allocated by an allocation policy. His later "law of least effort" states that when several routes reach the same goal, people gravitate to the least demanding [20]. Modern theory makes the allocation normative: the *expected value of control* weighs the payoff of engaging control against its intrinsic cost [36], and resource-rational analysis models cognition as optimal use of bounded computation [27]. One strand of this literature, ego depletion, failed a large multi-laboratory registered replication [16]; we therefore rely only on the cost–benefit formulation, which does not depend on it.

2.5 Noise and decision hygiene

Noise [23] separates *bias* (systematic error) from *noise* (unwanted variability), including occasion noise — the same judge answering differently on different occasions — and recommends *decision hygiene*: collect independent judgments before they can influence each other, then aggregate. Averaging independent estimates from one person improves accuracy, the "crowd within" [44].

3 Machine Instantiation

Table 2 contrasts the two systems. Jev's design reads like a specification of an idealised System 1: parallel typed judgments, no generation, and a confidence trained to be *calibrated* — a property human System 1 lacks. Qwen3-8B's thinking mode is serial, effortful in tokens and latency, and adjustable in depth, with performance reported to scale with the reasoning budget [33]. Two recent findings on reasoning models echo the psychology: models *overthink* easy problems [7], and on some tasks accuracy *falls* as reasoning lengthens — inverse scaling in test-time compute [13]. Related AI work includes the SOFAI architecture of fast and slow solvers under a metacognitive module [4], cost-aware LLM cascades [6], selective classification with risk guarantees [14], threshold calibration as multiple testing [2], difficulty-adaptive test-time compute [38], and self-consistency sampling [46]. In a companion study on tree

Table 2: The two systems (vendor documentation [17, 18, 33, 34]).

	System 1: Jev	System 2: Qwen3-8B
Output	typed answer; choice: per-option probabilities and a confidence; yes/no: $P(\text{yes})$ only	free text; answer parsed from final line
Generation	none	up to 32k tokens (thinking)
Latency, cost	$\sim$0.5 s; \$0.042 per million input tokens	GPU-bound; grows with budget B
Confidence	trained for calibration	verbalised confidence often overconfident [47]
Weak spots	numbers, dates, adversarial text	overthinking, inverse scaling [7, 13]

ensembles [26] the fast/slow mapping failed as a speed claim; here the cost gap is real.

4 A Formal Correspondence

Let a query consist of a text state, a typed question with options $\mathscr{A}$ ($|\mathscr{A}| = K$) and gold answer y. System 1 returns a_1, a probability vector p and a confidence c. We take c to be the confidence Jev itself reports: the `confidence` field for choice questions, and $\max(p_{\text{yes}}, 1-p_{\text{yes}})$ for yes/no questions, which return only p_{yes} [18]. The reported confidence need not equal $\max_k p_k$ (the vendor's example returns 0.82 with a top probability of 0.85), so the protocol tests the calibration of c itself and repeats the confidence-based analyses with $\max_k p_k$ (for yes/no questions the two coincide). System 2 with thinking budget B returns a_2 (B=0 is non-thinking mode). A controller chooses to *act* (release a_1), to *deliberate* with budget B (release a_2), or to fall back. Z denotes the set of deliberated queries. Table 4 lists the correspondences developed below, and Table 3 gives the notation in plain words.

5 Mathematics of the Interaction

Table 5 collects the assumptions behind the results of this section, each with its plain meaning, the results that need it, the test in the protocol that checks it, and what still holds if it fails. Two results, the escalation identity and the paired decomposition, make no assumption about either model: they are accounting identities, true on any data (the first is stated for 0–1 loss, A7, but has a version for any loss).

5.1 Conflict and override: the escalation identity

PROPOSITION 5.1 (ESCALATION IDENTITY). *Under 0–1 loss, the gain in overall accuracy from releasing a_2 instead of a_1 on the queries in Z is*

$$G(Z) = M_Z\,(W_2 - W_1), \qquad M_Z = P(x{\in}Z,\ a_1{\neq}a_2),$$

where $W_k = P(a_k{=}y \mid x{\in}Z, a_1{\neq}a_2)$. For $K{=}2$, $G(Z) = M_Z(2W_2 - 1)$.

PROOF. $G(Z) = \mathbb{E}[\mathbf{1}_Z(\mathbf{1}\{a_2{=}y\} - \mathbf{1}\{a_1{=}y\})]$; the integrand vanishes when $a_1{=}a_2$. Condition on disagreement. For $K{=}2$ exactly one of two disagreeing answers is correct. □

Table 3: Notation in plain words.

Symbol	Meaning
a_1, c	Jev's answer and how sure Jev says it is
a_2, B	Qwen3-8B's answer after thinking for B tokens
τ	act threshold: Jev's answer is released unchecked when $c \geq \tau$
M_Z	how often the two models disagree (countable without an answer key)
W_1, W_2	how often Jev, or Qwen3-8B, is right when they disagree (needs labels)
$d = 2(1{-}c)$	difficulty for yes/no: 0 when Jev is certain, 1 at a coin flip
$s(c)$	how much accuracy there is to gain by thinking at confidence c
$h(B)$	share of that gain that B tokens capture ($h(0) = 0$)
λ	price of one thinking token, in accuracy points
r	chance that decisive thinking talks Qwen3-8B out of a correct answer
γ, β	shape of the fixability $1 - d^\gamma$; token scale of h
ε, δ	tolerated act-band error; allowed chance that a certificate is wrong
ε_{g}	give-up tolerance: fall back when the error expected after thinking exceeds it
α	anchoring rate (Cor. 5.5); miscoverage of a conformal set (Prop. 5.10)
q	share of act-band answers re-checked in shadow for monitoring

COROLLARY 5.2 (WHERE VERSUS WHETHER). *M_Z is computable on unlabelled traffic; $W_2 - W_1$ requires labels.*

Interpretation. The identity separates De Neys's conflict detection from Pennycook's override. M_Z is the machine analogue of Stage 2: it registers that the two systems disagree, and it can be measured without knowing who is right. $W_2 - W_1$ is the quality of Stage 3 decoupling. A deliberation that ends in agreement — System 2 endorsing System 1, Kahneman's lazy controller at work — contributes exactly zero, however much it cost.

In plain words. Thinking pays only on queries where the two models end up disagreeing, and only if Qwen3-8B is right more often than Jev on those queries. For example (illustrative numbers), if they disagree on 12% of queries and Qwen3-8B wins 70% of those yes/no disagreements, deliberation adds $0.12 \times (0.70 - 0.30) = 4.8$ accuracy points. The first number can be counted on live traffic with no answer key; the second needs labels.

5.2 Trustworthy intuition: the act band

PROPOSITION 5.3 (ACT-BAND GUARANTEE). *If System 1 is calibrated, $P(a_1{=}y \mid c) = c$, then for any threshold τ, $P(a_1{\neq}y \mid c \geq \tau) = \mathbb{E}[1 - c \mid c \geq \tau] \leq 1 - \tau$.*

PROOF. The band is a function of c, so calibration holds on it; apply the tower property. □

Remark 1. Calibration is an aggregate property [17], and Kahneman and Klein's conditions for valid intuition [22] have a direct statistical reading: the act band is trustworthy only on traffic resembling the feedback on which it was certified. Given n labelled items, scan τ from high to low and stop at the first threshold whose Clopper–Pearson upper bound on act-band error exceeds ε. Fixed-sequence

Table 4: Kahneman's constructs and their formal counterparts in a two-model decision system.

Psychological construct	Formal object	Result and prediction
Feeling of rightness [41]; cognitive ease	calibrated confidence c	Prop. 5.3: error of acting on $c \geq \tau$ is at most $1-\tau$
Trustworthy intuition requires valid cues and feedback [22]	calibration on exchangeable traffic; labelled feedback set	Remark 1: a few hundred labels suffice to certify the act band; fails out of distribution
WYSIATI, illusion of validity [20]	confident errors, $P(a_1 \neq y \mid c \geq \tau) > 1-\tau$	act-band failure on adversarial or lure items (H1′, H8)
Conflict detection [10]; Stage 2 [32]	disagreement mass $M_Z = P(Z, a_1 \neq a_2)$	Prop. 5.1: observable without labels
Override / decoupling [32]	win margin $W_2 - W_1$ on disagreements	Prop. 5.1: deliberation helps iff System 2 is right more often than System 1 on disagreements
Anchoring [43]; rationalisation [32]	paired transitions blind $\rightarrow$ informed	Prop. 5.4: anchor, reactance and drift terms, exact
Decision hygiene [23]; consider the opposite [28]	blind escalation; blind-then-reconcile	Cor. 5.5: pure anchoring scales the benefit by $1-\alpha$
Effort as limited capacity [19]; EVC [36]; resource rationality [27]	$\max_B s(c)h(B) - \lambda B$	Props. 5.6–5.7: act/deliberate/give-up bands
Law of least effort [20]	zero budget above a confidence threshold	act band edge depends only on $\rho^\star = \sup h(B)/B$
Overthinking, rumination	flip rate r; inverse scaling	budget capped at the peak of h (Prop. 5.7a)
Knowing when to give up [1]	fallback when post-deliberation error $> \varepsilon_g$	Props. 5.9–5.10: Bayes action, conformal sets
Occasion noise; crowd within [23, 44]	majority of k System-2 samples	Prop. 5.11: voting corrects noise, entrenches bias

Table 5: Assumptions in plain words: where each is used, how it is checked, and what happens if it fails.

Assumption	In plain words	Used by	Checked by	If it fails
A1 Calibration	When Jev says 90%, it is right about 9 times in 10	Prop. 5.3; Prop. 5.9 (for every option)	reliability diagram and ECE on the calibration split; H1, H1′	use the certificate of Remark 1, which needs only A2
A2 Exchangeability	Tomorrow's queries look like the labelled ones	Rem. 1, Props. 5.3, 5.10	label-free drift alarm on M; audited errors with a sequential test (Sec. 9)	recertify τ and the conformal quantile; widen the deliberate band meanwhile
A3 Separable returns	Benefit of thinking = what there is to gain at this confidence $\times$ the share of it B tokens capture	Props. 5.6–5.7	budget sweep by task and confidence bin; H4	condition the budget table on task type; a pooled table kept 70% of the value in our numerical case
A4 Saturating returns	Each extra token helps a little less than the last	Prop. 5.6 only	same sweep	use the nonparametric table (P2); Prop. 5.7 needs no shape
A5 Error model	Thinking is decisive with a chance that grows with B and shrinks near a coin flip; decisive thinking fixes a wrong answer, and breaks a right one with chance r	Prop. 5.6 only	fix and break rates by band in the sweep	Props. 5.7–5.8 still hold
A6 Independent runs	Repeated Qwen3-8B runs make unrelated mistakes	Prop. 5.11	noise audit; H9	fewer effective votes; gains shrink and can turn negative even when per-run accuracy exceeds $\frac{1}{2}$
A7 0–1 loss	Every wrong answer costs the same	Prop. 5.1 as stated	—	identity holds with $W_2 - W_1$ replaced by the mean loss saved on disagreements
A8 Pure anchoring	The hint only makes Qwen3-8B copy Jev, at a rate unrelated to who is right	Cor. 5.5 only	transition shares; H3	Prop. 5.4 still holds exactly

testing then bounds act-band error by ε with probability $\geq 1-\delta$ under exchangeability, whether or not Jev is calibrated [2], provided the order of the tests does not depend on the labelled sample. A band with fewer than $\lceil \ln\delta / \ln(1-\varepsilon) \rceil$ items (59 for $\varepsilon = \delta = 0.05$) cannot pass even with no errors, so we start the scan at the first threshold whose band is expected to reach that size, judged from the confidences of *unlabelled* queries rather than of the labelled sample. WYSIATI predicts where this breaks: text engineered to be coherent but misleading (prompt injection, distractors, CRT-style lures) can make a System-1 model confidently wrong, and no threshold on its own confidence will catch that.

In plain words. If Jev's "90%" really means 90%, then releasing its answers whenever it is at least 90% sure gives at most 10% errors among the released answers. Because "really means" can fail, Remark 1 gives a check that does not rely on it: label a sample of past queries and lower the threshold only while the error among released answers is provably below the tolerance. The certificate covers only traffic that resembles the labelled sample.

5.3 Anchoring and rationalisation: paired transitions

Showing System 2 the pair (a_1, c) can reshape its entire chain of thought, not merely swap its final answer. We make no assumption about that mechanism and compare outputs in a *paired* design: on each escalated query System 2 answers blind (b) and informed (i).

PROPOSITION 5.4 (PAIRED TRANSITION DECOMPOSITION). *Partition escalated queries into* stable *($i{=}b$),* anchor *($b{\neq}a_1$, $i{=}a_1$),* reactance *($b{=}a_1$, $i{\neq}a_1$) and* drift *($b{\neq}a_1$, $i \notin \{a_1, b\}$, only for $K{>}2$). Then, with $P(T)$ the probability that a query is escalated and of type T,*

$$G^{\text{inf}} - G^{\text{blind}} = \textstyle\sum_T P(T)\big(P(i{=}y \mid T) - P(b{=}y \mid T)\big),$$

with a zero stable term, for any effect of the hint on the reasoning trace.

PROOF. $G^{\text{inf}} - G^{\text{blind}} = \mathbb{E}[\mathbf{1}_Z(\mathbf{1}\{i{=}y\} - \mathbf{1}\{b{=}y\})]$ and the four types partition Z. □

COROLLARY 5.5 (PURE ANCHORING). *If only anchor transitions occur, at a rate α independent of correctness given blind disagreement, then $G^{\text{inf}} = (1-\alpha)G^{\text{blind}}$.*

Interpretation. Anchor transitions are Stage-3 *rationalisation* in Pennycook et al.'s sense: the analytic process arrives where the initial response already was. Reactance, where a hint pushes System 2 away from a correct a_1, and drift, where it derails reasoning to a third option, are separate terms rather than unmodelled noise. Corollary 5.5 is the formal case for Kahneman et al.'s decision hygiene: obtain the independent judgment first. The blind-then-reconcile pattern (Sec. 6) does exactly that, confining any anchoring to a short, auditable "consider the opposite" step [28].

In plain words. Shown Jev's answer, Qwen3-8B may simply copy it. To measure this, ask Qwen3-8B twice on the same query, without and with Jev's answer, and sort each pair: unchanged, switched to Jev's answer (anchoring), switched away from it (reactance), or switched to a third option (drift). The accuracy difference between the two versions is exactly the sum of these pieces. If Qwen3-8B copies Jev on a share α of its disagreements, it keeps only $1-\alpha$ of the benefit of thinking.

5.4 Effort as a resource: how much to think

Kahneman's allocation policy [19] and the expected value of control [36] weigh the benefit of control against its cost. In our setting the benefit of thinking budget B on a query with confidence c is $s(c)h(B)$ and the cost is λB.

PROPOSITION 5.6 (SATURATING RETURNS). *Let $K{=}2$ and $d = 2(1-c)$. Suppose deliberation is* decisive *with probability $(1-d^\gamma)h(B)$, $h(B) = 1 - e^{-B/\beta}$; decisive deliberation fixes a System-1 error, and overturns a correct System-1 answer with probability r. The net value $s(c)h(B) - \lambda B$, where $s(c) = [(1-c) - rc](1-d^\gamma)$, is maximised by $B^* = \beta \ln(s/\lambda\beta)$ if $s > \lambda\beta$ and $B^* = 0$ otherwise. No thinking is bought for $c \geq (1-\lambda\beta)/(1+r)$; $s(c) \to 0$ as $c \to \frac{1}{2}$; and B^* peaks at intermediate confidence.*

PROOF. An error is fixed with probability $(1-d^\gamma)h(B)$ and a correct answer overturned with probability $r(1-d^\gamma)h(B)$, so the expected gain is $s(c)h(B)$. The objective is concave in B; solve $se^{-B/\beta}/\beta = \lambda$. Since $s \leq 1 - c(1+r)$, $s > \lambda\beta$ requires $c < (1 - \lambda\beta)/(1+r)$. □

The three regions are Kahneman's: the law of least effort (act when confident), effortful engagement at intermediate confidence, and a metacognitive decision to give up [1] on queries that are intrinsically ambiguous, where System 1's error is highest but deliberation cannot help. The overthinking ratio r sets the upper edge.

In plain words. $s(c)$ is what there is to gain at confidence c: the chance that Jev is wrong, $1-c$, minus the risk that thinking talks Qwen3-8B out of a right answer, rc, scaled down by $1-d^\gamma$ for items so close to a coin flip that no amount of thinking settles them. $h(B)$ is the share of that gain that B tokens capture, and λB is what the tokens cost. The best budget grows only with the logarithm of what is at stake: doubling s adds $\beta \ln 2 \approx 0.7\beta$ tokens, about 700 at $\beta = 1024$. With the illustrative values used below ($r = 0.1$, $\lambda\beta = 0.01$, so 1,024 tokens cost one accuracy point), no thinking is bought above $c = 0.99/1.1 = 0.90$.

Real thinking returns need not be smooth: they can flatten almost at once, require a threshold of reasoning before any benefit, or decline as reasoning lengthens [7, 13]. We keep only separability.

PROPOSITION 5.7 (SHAPE-FREE STRUCTURE). *Let $h \geq 0$ be any function with $h(0) = 0$ that attains its maximum, $\bar{B}$ the smallest maximiser of h, and $\rho^\star = \sup_{B>0} h(B)/B$. Then: (a) every optimal budget lies in $[0, \bar{B}]$; (b) the largest optimal budget is nondecreasing in $s(c)$; (c) $B^*(c) > 0$ iff $s(c) > \lambda/\rho^\star$, so with s unimodal the deliberate region is an interval and the curve affects its edges only through $\rho^\star$; (d) if $h(B)/B$ has a unique peak at $B_{\tan} > 0$, no positive budget below $B_{\tan}$ is optimal.*

PROOF. (a) For $s \leq 0$ the optimum is $B = 0$; for $s > 0$, beyond $\bar{B}$ returns do not rise and cost does. (b) For $s > 0$, maximising $sh(B) - \lambda B$ is maximising $h(B) + tB$ with $t = -\lambda/s$, which rises with s; $h(B) + tB$ has increasing differences in (B, t) for any h, so monotone comparative statics [30, 42] applies; for $s \leq 0$ the optimum is 0. (c) $B{>}0$ beats $B{=}0$ iff $sh(B) > \lambda B$ for some $B > 0$. (d) At an optimum $0 < B < B_{\tan}$, $sh(B) \geq \lambda B$, so $s > 0$ and $h(B)/B \geq \lambda/s$; moving to $B_{\tan}$ adds value $s[h(B_{\tan}) - h(B)] > (B_{\tan} - B)\,sh(B)/B \geq \lambda(B_{\tan} - B)$, more than its extra cost. □

PROPOSITION 5.8 (MISSPECIFICATION REGRET). *If $\hat{g}$ estimates the net accuracy gain g with $\sup_B |g - \hat{g}| \leq \eta$, the plug-in budget $\arg\max_B \hat{g}(B) - \lambda B$ loses at most 2η.*

PROOF. $V(B^*) - V(\hat{B}) \leq [V(B^*) - \hat{V}(B^*)] + [\hat{V}(\hat{B}) - V(\hat{B})] \leq 2\eta$. □

In plain words. Whatever the shape of the thinking curve, three things hold: never think past the point where more tokens stop helping; think more when more is at stake; and the confidence at which thinking starts to pay depends on the curve only through its best accuracy-per-token ratio $\rho^\star$. If returns need a running start, small budgets are wasted: think enough or not at all. And a budget table estimated from data with error at most η loses at most 2η against the best possible table.

Behaviour under realistic scaling. Table 6 and Fig. 1 evaluate the exact optimum for five return curves spanning reported behaviour, using the parameters of Prop. 5.6 ($r{=}0.1$, $\gamma{=}2$, $\lambda\beta{=}0.01$,

Table 6: Optimal budgets under five return curves (numerical; model properties).

$h(B)$	act $c\geq$	give up $c\leq$	peak B^*	min $B^*>0$	kept
saturating	0.90	0.57	2.76k	57	100%
log-linear	0.90	0.59	2.62k	10	97%
early-flat	0.91	0.58	0.55k	8	71%
threshold	0.85	0.57	5.52k	3.74k	45%
inverted-U	0.90	0.56	2.07k	10	97%

"kept": value of the saturating closed-form budget relative to the true optimum.

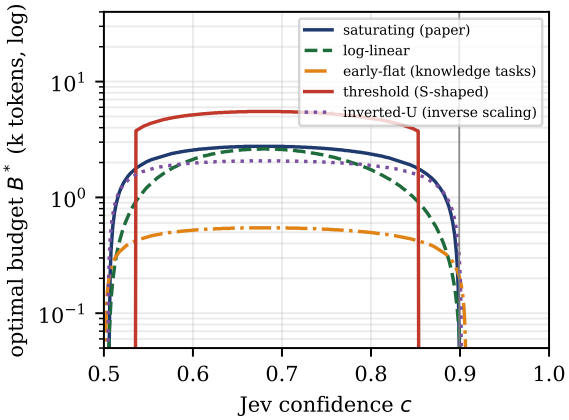


Figure 1: Optimal thinking budget $B^*(c)$ under five return curves (log scale). Band edges are nearly invariant; magnitudes and the minimum effective budget are not.

β=1024, give-up tolerance ε_g=0.35). These are properties of the model, not measurements. The bands barely move: acting begins at $c \in [0.85, 0.91]$ and giving up ends at $c \in [0.56, 0.59]$ for every curve. Magnitudes move tenfold. Under threshold returns a minimum effective budget of 3.7k tokens appears, so the policy becomes all-or-nothing — a cascade. A closed form fitted for saturating returns keeps 97–100% of the achievable value under three curves but only 71% (early-flat) and 45% (threshold). Decisions about *whether* to think are robust; decisions about *how much* are not. When curves differ by task, a single pooled policy keeps 70% of the value of a task-conditioned one; Jev can supply the task label through an auxiliary typed question answered in the same call.

5.5 Giving up without a human

Many real-time services cannot route hopeless queries to a person. Jev returns a probability for every option, which supports two principled fallbacks.

PROPOSITION 5.9 (CALIBRATED BAYES ACTION). *If $P(y{=}k \mid p) = p_k$ for all k, then for any loss $L(a,y)$ over an action set that may include a conservative no-op ("hold", "safe queue"), the rule $a^*(p) = \arg\min_a \sum_k p_k L(a,k)$ minimises expected loss among all rules that depend on the query only through p.*

PROOF. $\mathbb{E}[L(\delta(p),y)] = \mathbb{E}[\sum_k p_k L(\delta(p),k)]$ by calibration; minimise pointwise. □

PROPOSITION 5.10 (CONFORMAL ANSWER SETS). *With n exchangeable labelled items and $\hat{q}$ the $\lceil (n{+}1)(1{-}\alpha) \rceil$-th smallest score $1 - p_y$ ($\hat{q} = \infty$ if that rank exceeds n), the set $C(x) = \{k : 1 - p_k(x) \leq \hat{q}\}$ satisfies $P(y \in C(x)) \geq 1 - \alpha$ for a new exchangeable query [35].*

Prop. 5.9 needs calibration of the *full* vector, which is stronger than confidence calibration and is tested in the protocol. It also marks a deliberate departure from human System 2, whose choices under risk show loss aversion [24]: the machine fallback should encode the organisation's actual losses, not reproduce a descriptive bias. The fallback order is: a human if one is available; otherwise information acquisition (retrieval, a clarifying question) when affordable; otherwise the Bayes action, or a minimax action over the conformal set.

In plain words. With no human available, the Bayes action picks the option, or a safe "hold", with the lowest expected cost under Jev's probabilities; it is the best such rule only if those probabilities are calibrated for every option. A conformal set instead returns every option that cannot be ruled out; with $\alpha = 0.1$ it contains the right answer at least 90% of the time, with no calibration assumption.

5.6 Noise and the crowd within

Sampling System 2 several times and taking the majority (self-consistency [46]) is the machine version of the crowd within [44]. Kahneman et al.'s distinction between bias and noise determines when it helps.

PROPOSITION 5.11 (AGGREGATION CORRECTS NOISE, ENTRENCHES BIAS). *Let $K{=}2$ and let System 2 answer query i correctly with probability p_i on each of $k \geq 3$ (odd) independent runs. The majority is correct with probability $M_k(p_i) = P(\mathrm{Bin}(k, p_i) > k/2)$, and $M_k(p) - p$ has the sign of $p - \frac{1}{2}$, with $M_k(p) \to \mathbf{1}\{p > \frac{1}{2}\}$ as $k \to \infty$. Hence the expected gain from voting, $\mathbb{E}_i[M_k(p_i) - p_i]$, is positive on queries where System 2 is noisy but right on average and negative where it is systematically wrong.*

PROOF. $M_k(p) + M_k(1-p) = 1$, so $M_k(\frac{1}{2}) = \frac{1}{2}$; M_k is increasing, with $M_k'(p) = k\binom{k-1}{(k-1)/2}[p(1-p)]^{(k-1)/2}$ exceeding 1 near $\frac{1}{2}$ and vanishing at the ends. Hence $M_k(p) - p$ has a single interior root at $\frac{1}{2}$ on $(0,1)$ and takes the sign of $p - \frac{1}{2}$; the limit is the law of large numbers. This is the per-query form of Condorcet's jury theorem [15]. □

Fig. 2 plots the effect. Voting is decision hygiene against occasion noise, but on items where System 2 has a systematic misconception — the machine analogue of a biased judge — more samples make the error more certain. Samples from one model are also correlated, and correlation does more than shrink the effective k: if accuracy varies across runs of the same query, voting can lose accuracy even when the average run is right more often than not. The protocol therefore runs a *noise audit*: the rate at which repeated runs on the same query disagree.

In plain words. Asking Qwen3-8B several times and taking the majority helps on queries it usually gets right and hurts on queries

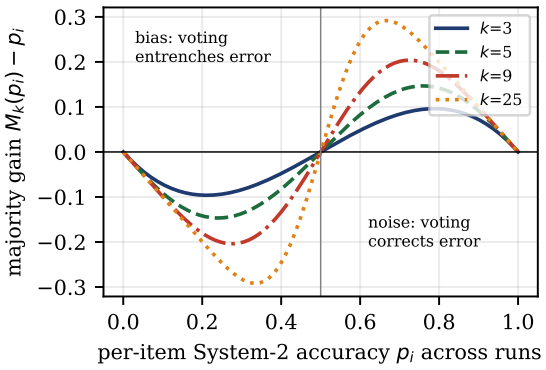


Figure 2: Prop. 5.11: gain from majority voting over k System-2 runs as a function of per-query accuracy. Voting corrects noise ($p_i > \frac{1}{2}$) and entrenches bias ($p_i < \frac{1}{2}$).

it usually gets wrong: voting removes random slips, not systematic misconceptions.

6 Interaction Patterns

Dual-process theories disagree about architecture. *Default-interventionist* accounts have System 1 answer first and System 2 intervene when needed [11]; *parallel-competitive* accounts run both and resolve conflict [37]. Both have machine counterparts (Fig. 3). **P1 Cascade** (default-interventionist): deliberate at a fixed budget when $c < \tau$, with τ certified as in Remark 1. **P2 Budgeted deliberation**: deliberate with a task-conditioned budget estimated nonparametrically on a budget grid (Prop. 5.8 bounds its regret); the closed form of Prop. 5.6 is a baseline. **P3 Disagreement probe** (parallel-competitive): run Qwen3-8B in non-thinking mode, which costs tens of tokens, alongside Jev, and think only on disagreement or when $c < \tau$. This estimates the label-free conflict signal M directly and can catch confident System-1 errors that a confidence threshold misses — De Neys's conflict detection without low confidence. **P4 Jev as verifier**: when Qwen3-8B overturns Jev, Jev judges whether the new answer is supported by the passage and reasoning; if not, the override is reverted. This inverts the usual direction and targets Stage-3 rationalisation by System 2 itself. **P5 Blind-then-reconcile**: System 2 reasons blind, then sees (a_1, c) in a short second turn and may revise only on finding a concrete error — decision hygiene as an interaction protocol.

6.1 When each model checks the other

By default Qwen3-8B does *not* check Jev. Jev answers every query; the controller reads Jev's confidence; and only queries below the certified threshold τ that the budget table allots thinking go to Qwen3-8B, which answers *blind*, without seeing Jev's answer. That independent answer, compared with Jev's, is the check: agreement releases the common answer, and disagreement releases Qwen3-8B's. Above τ, Jev's answer is released unchecked, except that a random share q of these answers is re-answered by Qwen3-8B in shadow, for monitoring only. Three options change the default. The probe (P3) extends a cheap, non-thinking check to every query, including confident ones. Reconciliation (P5) shows Jev's answer to Qwen3-8B, but only after its blind answer exists. The verifier (P4) reverses the direction: whenever Qwen3-8B overturns Jev, Jev checks Qwen3-8B. The *informed* arm, in which Qwen3-8B sees (a_1, c) from the start, exists only to measure anchoring in the evaluation and is not a deployment pattern. Table 7, Fig. 3 and Algorithm 1 use the same step numbers.

Table 7: Who checks whom, and when (one query).

#	Who checks whom	When	Sees the other's answer?
1	Jev answers	every query	—
2	Qwen checks Jev (probe, P3)	every query, if enabled; no thinking	no
3	nobody: a_1 released	$c \geq \tau$ and no probe conflict	—
3a	Qwen re-checks Jev (audit)	random share q of step 3; output unchanged	no
4	Qwen checks Jev (P1, P2)	$c < \tau$, or probe conflict; budget B	no
5	Qwen reconsiders (P5)	after step 4, if enabled	yes: (a_1, c)
6	Jev checks Qwen (P4)	after steps 4–5, only if Qwen's answer $\neq a_1$	yes: answer and reasoning
—	fallback	$B = 0$ and $1 - c > \varepsilon_g$, or expected post-thinking error $> \varepsilon_g$	—

Algorithm 1: Dual-process controller (one query). Step numbers match Table 7 and Fig. 3.

Input: query x; certified threshold τ; budget table $\hat{B}(c, \text{task})$; give-up tolerance ε_g; budgets B_{conf}, B_a; verifier level κ; audit share q; loss L; flags *probe*, *reconcile*, *verify*

$(a_1, p, c, \text{task}) \leftarrow \text{Jev}(x)$ `// 1: one call`
conflict $\leftarrow$ false
if *probe* **then** $a_0 \leftarrow \text{Qwen}(x, 0)$ blind; conflict $\leftarrow [a_0 \neq a_1]$ `// 2`
if $c \geq \tau$ ***and not*** *conflict* **then return** $\text{Act}(a_1)$ `// 3: act`
$B \leftarrow \hat{B}(c, \text{task})$; **if** *conflict* **then** $B \leftarrow \max(B, B_{\text{conf}})$
if $B = 0$ **then return** $\text{Act}(a_1)$ if $1 - c \leq \varepsilon_g$, else $\text{Fallback}(p)$
if *GiveUp*(c, B) **then return** $\text{Fallback}(p)$ `// 3: give up`
$b \leftarrow \text{Qwen}(x, B)$ blind `// 4: Qwen checks Jev`
if *reconcile* **then** $b \leftarrow \text{Reconcile}(b; a_1, c)$ `// 5`
if *verify* ***and*** $b \neq a_1$ **then**
  if *Jev rates b unsupported with prob.* $\geq \kappa$ **then return** a_1 `// 6`
return b

Act(a_1): with probability q, log a blind $\text{Qwen}(x, B_a)$ run for audit (3a); release a_1.
GiveUp(c, B): the error expected after thinking (Prop. 5.6 or the sweep) exceeds ε_g; for a fixed cascade, c below a give-up threshold.
Fallback(p): human if available; else acquire information if affordable and restart; else $\arg\min_{a \in \mathcal{A} \cup \{\text{hold}\}} \sum_k p_k L(a, k)$ or a minimax action over the conformal set.

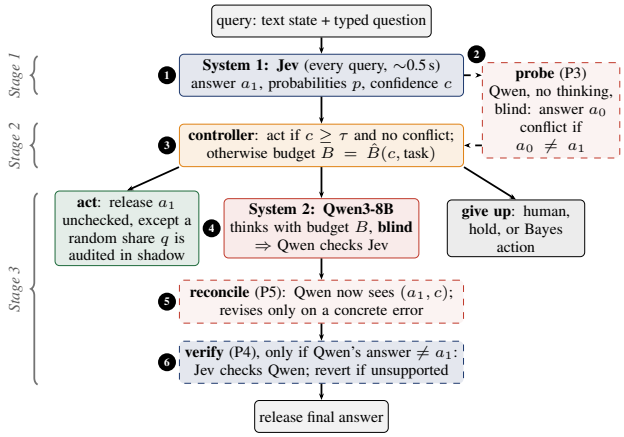


Figure 3: Who checks whom, mapped onto Pennycook et al.'s three-stage model [32]. Stage 1: Jev's answer and feeling of rightness. Stage 2: conflict monitoring sets the budget. Stage 3: System 2 decouples or rationalises, audited by Jev. Solid: default path; dashed: optional patterns. Numbers are the steps of Table 7.

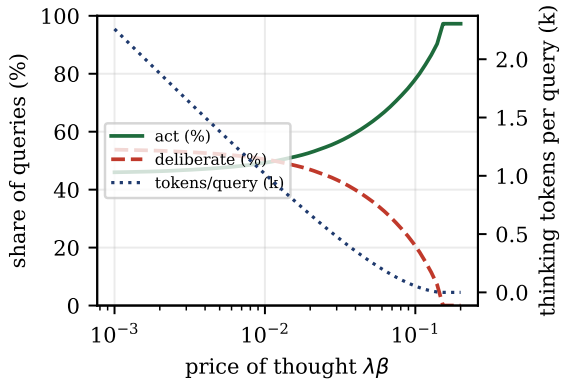


Figure 4: Price of thought (worked example): as thinking becomes more expensive, queries move from deliberation to acting and tokens per query fall.

7 Applications in Service and Software Systems

The controller fits any service in which many decisions are cheap to make, a minority deserve deliberation, and mistakes have asymmetric costs. Table 8 summarises seven domains within the scope of this conference.

Service operations. In a help desk, Jev answers routing questions for every ticket and Qwen3-8B drafts plans only for tickets in the deliberate band. The loss matrix is explicit: a mis-routed urgent ticket costs far more than a short delay, so the Bayes fallback (Prop. 5.9) will often prefer the "priority queue" no-op to a confident but uncertain category. *Software engineering.* Pull-request risk routing, CI-failure triage and incident paging are high-volume typed decisions; community tools already route pull requests and model tiers with Jev [17]. Deliberation over diffs and logs is expensive and prone to fluent but wrong root-cause stories, which the verifier (P4) targets. *Requirements engineering.* Classifying requirements and flagging ambiguity is a typed task; conformal sets expose requirements whose classification is genuinely uncertain, which should be clarified with stakeholders rather than guessed. *Logistics.* Exception notices for shipments and vessel calls (delay causes, customs holds, dangerous-goods declarations) arrive at volume; the natural fallback is to hold for inspection. The WYSIATI risk is concrete: a well-formed but false notice can make System 1 confidently wrong, which is why the probe (P3) runs in parallel. *Consumer behaviour.* Consumer-facing services judge behaviour at volume: whether a review is genuine, whether a message signals intent to cancel, whether a return request fits the purchase history, which category a complaint belongs to. Jev answers these typed questions for every event and acts on the confident majority; Qwen3-8B reads the middle band — a complaint that mixes issues, a review whose text and rating disagree — and drafts the reply or the retention offer; and the fallback is a human agent or a Bayes action whose loss treats refusing a genuine customer as costlier than a short hold. The domain is also one Kahneman's work helped shape: consumers' own choices show reference dependence and loss aversion [24, 39], and choice architecture steers them [40]. A controller that reads consumer behaviour should therefore be audited for bias and noise across customer groups, and it should not be used to exploit the fast, System-1 choices it detects, as dark patterns do [29]. A shift in the label-free disagreement rate M can flag a change in behaviour, such as a new fraud pattern or a surge of complaints, before labelled outcomes arrive. *E-government.* For decisions affecting rights and entitlements, human review must stay first in the fallback order regardless of cost, and disparities should be audited separately as bias (systematic, by group) and noise (inconsistency) in Kahneman et al.'s sense. *Agentic software.* Judging whether each tool result is relevant or contains an injected instruction before it enters an agent's context is a System-1 task at high frequency; deliberation is reserved for planning.

Worked example (illustrative parameters, not measurements). Take a binary triage service whose System-1 confidence is distributed as $c = 0.5 + 0.5U$, $U \sim \text{Beta}(3, 1)$, and use the saturating model of Prop. 5.6. Acting on System 1 alone yields accuracy 0.875. Thinking at 4,096 tokens on every query yields 0.895: the gain on uncertain queries is partly cancelled by overthinking on confident ones. Table 9 shows the budgeted controller across three price regimes. At $\lambda\beta$=0.01 it decides about half the queries fast, deliberates on the other half, gives up on 0.2%, and reaches 0.915 on decided queries at about 1,000 tokens per query — higher accuracy than thinking on everything at a quarter of the tokens. Raising the price of thought shifts queries from deliberation to acting, as the law of least effort predicts.

Fig. 4 traces the whole frontier: tokens per query fall from about 2,300 to zero as the price of thought rises, while accuracy on decided queries declines gracefully from 0.92 towards the System-1 level.

7.1 Deployment and quality assurance

Service levels. Jev runs on every query; Qwen3-8B only on the deliberate share, so tail latency is a mixture whose weight is read off Fig. 4, and capacity planning reduces to choosing the price

Table 8: Applications. Each row names System 1's typed questions, System 2's deliberative task, the dominant asymmetric loss, the fallback, and the dual-process risk to monitor.

Domain	System 1 (Jev) typed questions	System 2 (Qwen3-8B) task	Costly error	Fallback	Risk to monitor
Customer service / CRM	category; urgency; churn risk; refund eligibility	resolution plan for complex, multi-issue tickets	urgent ticket mis-routed	priority queue (hold)	anchoring on System-1 category
Software engineering	PR risk tier; test failure flaky vs. real; page on-call?	reason over diff or logs; propose root cause	missed security regression	block merge; page human	rationalised overrides (use P4)
Requirements engineering	functional vs. non-functional; ambiguous? testable?	rewrite ambiguous requirement; derive acceptance tests	ambiguity shipped to build	conformal set; ask stakeholder	overthinking clear items
Logistics / port operations	delay cause; customs hold; dangerous-goods flag	re-plan berth or route under constraints	unsafe cargo released	hold for inspection	WYSIATI on coherent but false notices
Consumer behaviour / e-commerce	review genuine? intent to cancel? return consistent with purchase history?	answer an ambiguous complaint; draft a retention offer	genuine customer refused or wrongly flagged	human agent; hold	bias vs. noise across customer groups
E-government services	eligibility pre-screen; document complete?	check against regulation text	wrongful denial	mandatory human review	bias vs. noise across groups
Agentic software	tool result relevant? injected instruction?	multi-step planning	context poisoning	drop result; stop agent	confident errors on adversarial text

Table 9: Worked example: shares, mean thinking tokens per query, and accuracy on decided queries under three prices of thought (illustrative).

Policy	act	deliberate	give up	tokens	accuracy
System 1 only	100%	—	—	0	0.875
Think on all, B=4096	—	100%	—	4,096	0.895
Budgeted, $\lambda\beta$=0.003	47%	53%	0.2%	1,658	0.918
Budgeted, $\lambda\beta$=0.01	49%	51%	0.2%	1,015	0.915
Budgeted, $\lambda\beta$=0.03	56%	43%	0.4%	478	0.906

of thought. *Failure handling.* When Jev errors or returns an out-of-schema answer, high-stakes services should fail *closed*, to the fallback rather than to a default route — the convention already adopted by community Jev routers. *Keeping intuition trustworthy.* Kahneman and Klein's feedback condition becomes an operational duty. The act band is certified on data from the past; to keep it valid, audit a random sample of act-band decisions and run Wald's sequential probability ratio test [45] of "error $\leq \varepsilon$" against "error $\geq \varepsilon + \Delta$", recertifying τ on rejection (Sec. 9 gives the audit volumes). Corollary 5.2 adds a label-free early warning: the disagreement rate M between Jev and the non-thinking probe can be tracked on all traffic, and a shift in M signals distribution drift before any labels arrive. *Versioning.* Thresholds, budget tables and conformal quantiles belong to a specific pair of model versions and must be recertified on any upgrade.

8 Pre-Specified Evaluation Protocol

8.1 Status of the evaluation

The protocol below has *not* been run. No call has been made to Jev or Qwen3-8B and no data have been collected, because the authoring environment had neither a Jev API key nor a GPU. What exists is: (i) the protocol, fixed in the file `PROTOCOL.md` and frozen with the code at the repository tag `protocol-v1.0` on 21 October 2026. It is *pre-specified* rather than pre-registered; it becomes a registration when it is deposited with a public registry, which we will do before the first model call, and any later change is logged with its date and reason. (ii) The code, which runs the whole protocol end to end on mock systems with known properties, including the analysis of every hypothesis except H8, which needs T3, and passes 42 unit tests, including request building and response parsing checked against the vendor's documented examples [18]. (iii) The numerical results of Secs. 5 and 7, which the repository regenerates from the stated parameters. What is needed is a Jev key and one GPU able to serve Qwen3-8B; Sec. 9 gives the token volumes. Every hypothesis will be reported whatever the outcome. Code, protocol, tests and the scripts that regenerate every number and figure in this paper are public under the MIT licence at `https://github.com/OWNER/jev-qwen-dual-process`; the frozen protocol is the tag `protocol-v1.0`.

8.2 Design

Tasks. T1: BoolQ [8] validation (3,270 yes/no items), a Jev strength. T2: GSM8K [9] test (1,319) recast as four-option choice with distractors fixed in advance, targeting Jev's documented weakness with arithmetic. T3: CRT-style lure items [12]: newly written isomorphs with fresh numbers and wording, to avoid contamination, in which the intuitive answer is wrong, plus distractor-laden items of the kind that produce inverse scaling [13]. The T3 items are not yet written; they and their hash will be committed before any model call (target $n = 200$). *Design.* 30% of each task is used for calibration (threshold certification; nonparametric budget curves on the grid $B \in \{0, 128, \ldots, 16\text{k}\}$, per task and confidence bin), 70% for test. All five patterns run; escalated items are run blind, informed and blind-then-reconcile in a paired design; a subset is run ten times for the noise audit: five runs vote, and five more estimate each item's accuracy independently, so that grouping items by accuracy does not make H9 circular. Qwen3-8B uses the vendor-recommended sampling settings [34]. Jev's model version is pinned in the request

Table 10: Testing in practice: four stages from replay to deployment.

Stage	Traffic and effect	What runs	Measured	Go / no-go
S0 Offline replay	benchmark or historical queries; no effect	Jev on all; Qwen3-8B blind at every budget, logged once	H1–H9; accuracy–token frontier for any τ and B by lookup	hypotheses as in Table 11; certify τ on the calibration split
S1 Shadow	live traffic; incumbent decides	full controller in parallel; outputs logged, not used	M by band (label-free); tokens; p95 latency; audited act-band errors	act-band error certified $\leq \varepsilon$ (59–361 labels for $\varepsilon = 5\%$); latency within the service level
S2 Canary	small share of traffic; fail-closed	controller decides; audit share q	sequential test on audited errors; reverted overrides; human escalations	test accepts ε; no rise in escalations or complaints
S3 Rollout	all traffic	controller; audit share q; drift alarm on M	sequential test continues; M by band; spend	rejection or a shift in M triggers recertification; recertify on any model upgrade

Table 11: Pre-specified hypotheses and falsification criteria.

	Hypothesis	Falsified if
H1	T1: act-band error $\leq \varepsilon$ at the certified τ	error CI above ε
H1′	T2: Jev overconfident: on $c \geq 0.9$, error exceeds the mean of $1-c$	error CI at or below the mean of $1-c$
H2	$W_2 > W_1$ in the middle band only; $W_2 \leq W_1$ in the top band	top-band $W_2 - W_1$ CI above 0
H3	Anchor share > reactance share; informed worse than blind; reconcile has lower anchor share than informed	any inequality reversed with CI excluding 0
H4	Nonparametric B^* is 0 in the top band, positive in a middle interval; curves differ by task	$B^* > 0$ in top band, or no task difference
H5	Probe disagreement predicts thinking-mode disagreement (AUROC > 0.7)	AUROC ≤ 0.7
H6	Verifier cuts harmful overrides $\geq$ 30%, keeps $\geq$ 80% of beneficial ones	either missed
H7	Conformal coverage $\geq 1-\alpha-0.02$; Bayes action beats always-act under the stated loss	either missed
H8	T3: Jev confidently wrong on lures; probe catches more lure errors than any confidence threshold	threshold catches as many
H9	Voting gains confined to items with $\hat{p}_i > \frac{1}{2}$; losses where $\hat{p}_i < \frac{1}{2}$ ($\hat{p}_i$ from independent runs)	gains where $\hat{p}_i < \frac{1}{2}$

and the returned version is logged; c is defined as in Sec. 4. *Measures.* Accuracy on decided queries; give-up rate; M, W_1, W_2 by confidence band; act-band error with Clopper–Pearson bounds; transition shares; classwise calibration; conformal coverage and set size; occasion noise and majority gain; tokens, latency and cost.

Code. The released code isolates API calls behind a Jev client, whose request and response handling follows the vendor's documented schema [18], and an OpenAI-compatible Qwen3-8B client. It implements Algorithm 1, every pattern, the nonparametric budget policy, threshold certification, the paired decomposition, the noise audit and both fallbacks, and it logs every model call so that any policy can later be replayed offline. Tested against mock systems with known properties, the identities of Props. 5.1 and 5.4 hold to machine precision, the nonparametric policy gives no thinking in the top band, positive budgets in the middle band and larger budgets for threshold-shaped returns, conformal coverage matches its target, the certificate is falsely issued less often than δ, and majority voting gains shrink when runs are correlated. These checks establish only that the code computes what the paper defines; they say nothing about Jev or Qwen3-8B.

9 Testing the Approach in Practice

A benchmark tests the theory; a service must also know that the controller is safe on its own traffic. We propose four stages, each with a go/no-go rule (Table 10). The same controller code runs in all of them, and each stage needs labels only where Table 5 says an assumption must be checked.

Offline replay (S0). Because Qwen3-8B answers blind at every budget on the grid, the outcome of any cascade with threshold τ and budget B is a table lookup, so a single logged sweep traces the whole accuracy–token frontier; only the informed, reconcile and verify arms need extra calls. The protocol is S0 on public benchmarks; a service can run the same replay on its own historical queries.

Shadow mode (S1). The controller runs beside the incumbent process — for example, beside the agents who review flagged returns and complaints — and logs its decisions without acting on them. Two things are measured without labels: the disagreement rate M by confidence band, and tokens and tail latency. Labels are needed only for the act band, and few suffice. Certifying an act-band error of at most 5% with 95% confidence needs 59 labelled act-band items if Jev makes no errors there; if its true act-band error is 2.5%, 361 labels give an 80% chance of certification (allowing up to 11 errors).

Canary (S2) and monitoring (S3). A small share of live traffic is then decided by the controller, failing closed to the fallback on any schema error, while a random share q of act-band decisions is audited. Wald's test of "error $\leq 5\%$" against "error $\geq 8\%$" ($\alpha = \beta = 0.05$) needs on average 153 audits when the true error is 2.5%, 389 at 5%, 354 at 8% and 64 at 16%, and triggers recertification with probability 0.04 at 5% and 0.95 at 8% (simulated). At 10,000 act-band decisions a day, $q = 1\%$ yields 100 audits a day, so a decision arrives within days. Standard practice for staged rollout and production readiness applies throughout [5, 25].

Cost of the protocol. Generated tokens on Qwen3-8B are bounded by 79M for T1 and 32M for T2, assuming every test item is escalated,

every budget hits its cap and every call adds 64 answer tokens; the calibration sweep accounts for 33M and 13M of these. Dividing by a measured serving throughput gives GPU hours. Jev needs at most about 5,600 calls for T1 and costs under US$0.50 at the listed price even at 2,000 input tokens per call; output is free. All figures in this section are statistical calculations from the repository's planning script, not measurements.

10 Discussion

Where the analogy holds. The correspondence is structural rather than metaphorical: each construct in Table 4 is defined operationally and measured by a quantity the propositions constrain. The strongest parallels are conflict detection as label-free disagreement, effort allocation as expected value of control, and decision hygiene as blind escalation. *Where it breaks.* Human System 1 is not calibrated; Jev is trained to be, which is precisely why Prop. 5.3 is useful and also why its failure modes (WYSIATI on adversarial text) deserve dedicated tests. Human System 2 is not a separate agent with its own prior, whereas Qwen3-8B is, so "decoupling" here means disagreement between two models, not revision within one mind. A model's thinking tokens are not deliberation in the psychological sense, and we claim no equivalence of mechanism, only of decision structure. Finally, the loss-aversion contrast shows that the formalisation is normative: it tells a service how it *should* allocate deliberation, not how people do.

What the formalism predicts about people. The results are normative for machines but yield testable predictions for the psychology they borrow from. (i) The escalation identity implies that studies of reflection should report the rate of detected conflict and the success of override *separately*, since the benefit of reflection is their product; CRT scores conflate them. (ii) The budget results predict that rethinking time is *non-monotone* in the feeling of rightness: it should fall as confidence rises, as Thompson et al. found [41], but also fall again for items judged hopeless, where people give up [1]. (iii) Prop. 5.11 predicts that averaging a person's repeated answers helps on items they get right more often than not, and harms on items where a systematic lure dominates, such as CRT items; the crowd within should cure noise, not bias. (iv) Prop. 5.4 suggests that advice-taking studies [3] use paired designs that separate anchoring from reactance. *Limitations.* No empirical results: the protocol is specified and coded but not run (Sec. 8.1). Props. 5.6–5.7 assume separable returns in confidence and budget; Props. 5.3 and 5.10 require exchangeable traffic; Prop. 5.9 requires classwise calibration; Prop. 5.11 assumes independent runs, which samples from one model are not (Table 5 gives the fallback for each). How Jev computes its reported confidence is not documented, so whether c is calibrated is an empirical question (H1, H1′). The parameter values in Secs. 5 and 7 are illustrative and must be replaced by estimates from the sweep.

11 Conclusion

Kahneman's two systems can be given an exact mathematical form when System 1 is a calibrated decision model and System 2 a budgeted reasoning model. The value of deliberation is the rate of conflict times the quality of override; conflict is visible without labels, override quality is not. Calibrated confidence certifies when intuition can be trusted. Paired comparison measures anchoring and rationalisation without modelling them. Effort allocation produces act, deliberate and give-up bands for any shape of returns, but how much to think must be learned per task. Aggregating samples corrects noise and entrenches bias. These results give service and software engineers a principled way to decide when a fast system should wake a slow one — and a protocol and code that can prove them wrong.

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